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PROBLEMS SUBMIT CODE MY SUBMISSIONS STATUS HACKS ROOM STANDINGS CUSTOM INVOCATION

D. Tree and Queries

time limit per test: 1 second
memory limit per test: 256 megabytes

You have a rooted tree consisting of n vertices. Each vertex of the tree has some color. We will assume that the tree vertices are numbered by integers from 1 to n . Then we represent the color of vertex v as c_v . The tree root is a vertex with number 1.

In this problem you need to answer to m queries. Each query is described by two integers v_j, k_j . The answer to query v_j, k_j is the number of such colors of vertices x , that the subtree of vertex v_j contains at least k_j vertices of color x .

You can find the definition of a rooted tree by the following link:
[http://en.wikipedia.org/wiki/Tree_\(graph_theory\)](http://en.wikipedia.org/wiki/Tree_(graph_theory)).

Input

The first line contains two integers n and m ($2 \leq n \leq 10^5$; $1 \leq m \leq 10^5$). The next line contains a sequence of integers $c_1, c_2, \dots, c_n$ ($1 \leq c_i \leq 10^5$). The next $n - 1$ lines contain the edges of the tree. The i -th line contains the numbers a_i, b_i ($1 \leq a_i, b_i \leq n$; $a_i \neq b_i$) — the vertices connected by an edge of the tree.

Next m lines contain the queries. The j -th line contains two integers v_j, k_j ($1 \leq v_j \leq n$; $1 \leq k_j \leq 10^5$).

Output

Print m integers — the answers to the queries in the order the queries appear in the input.

Examples

input

Copy

8 5
1 2 2 3 3 2 3 3
1 2
1 5
2 3
2 4
5 6
5 7
5 8
1 2
1 3
1 4
2 3
5 3

output

Copy

2
2
1
0
1

input

Copy

4 1
1 2 3 4
1 2
2 3
3 4
1 1

output

Copy

4

Note

A subtree of vertex v in a rooted tree with root r is a set of vertices $\{u : dist(r, v) + dist(v, u) = dist(r, u)\}$. Where $dist(x, y)$ is the length (in edges) of the shortest path between vertices x and y .

Codeforces Round 221 (Div. 1)

Finished

Practice

→ Virtual participation

Virtual contest is a way to take part in past contest, as close as possible to participation on time. It is supported only ICPC mode for virtual contests. If you've seen these problems, a virtual contest is not for you - solve these problems in the archive. If you just want to solve some problem from a contest, a virtual contest is not for you - solve this problem in the archive. Never use someone else's code, read the tutorials or communicate with other person during a virtual contest.

Start virtual contest

→ Clone Contest to Mashup

You can clone this contest to a mashup.

Clone Contest

→ Submit?

Language: GNU G++23 14.2 (64 bit, ms)

Choose file: Choose File No file chosen

Submit

→ Last submissions

Submission	Time	Verdict
363970341	Feb/23/2026 02:39	Accepted
363970178	Feb/23/2026 02:35	Wrong answer on test 4
363970037	Feb/23/2026 02:30	Wrong answer on test 4

→ Problem tags

data structures dfs and similar trees

*2400

No tag edit access

→ Contest materials

Codeforces Round #221 (en)

Tutorial (en)